The Laporta Constants

Six Numbers at the Frontier of Physics and Mathematics

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Domain: QED / Number Theory / Multi-Loop Computation / Mathematical Constants

Status: Active — Attack 1 complete (6/6 null at 36 basis, 300 digits; 1/1 null at 66 basis, 400 digits). Attacks 2-6 pending.

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I. THE MOST PRECISELY MEASURED QUANTITY IN PHYSICS

The electron anomalous magnetic moment a_e has been measured to 13 significant digits. The measurement — one electron trapped in a Penning trap at Harvard, its spin precession compared to its orbital frequency — produces:

a_e(exp) = 0.00115965218059(13)

This number is predicted by quantum electrodynamics as a power series in the fine structure constant:

a_e = $A_1(\alpha/\pi) + A_2(\alpha/\pi)^2 + A_3(\alpha/\pi)^3 + A_4(\alpha/\pi)^4 + A_5(\alpha/\pi)^5 + \ldots$

The coefficients A_1 through A_5 are computed from Feynman diagrams. $A_1 = 1/2$ (Schwinger, 1948, one diagram). $A_2 = -0.3285\ldots$ (Petermann, Sommerfield, 1957, seven diagrams). A_3 involves 72 diagrams. A_4 involves 891 diagrams. A_5 involves 12,672 diagrams.

A_1 through A_3 are known analytically — every term expressed in closed form using rational numbers, π, $\zeta(3)$, $\zeta(5)$, ln(2), and polylogarithms. A_5 is known only numerically to moderate precision.

A_4 sits between. In 2017, Stefano Laporta published the complete four-loop calculation. He evaluated all 891 diagrams numerically to extraordinary precision — 4800+ digits. The calculation reduced the diagrams to master integrals using integration-by-parts identities. Most master integrals were evaluated analytically. Six could not be.

Those six master integrals — labeled C81a, C81b, C81c, C83a, C83b, C83c from their topology numbers in Laporta's classification — are known to 4925 digits but have no known closed form. For eight years, the multi-loop community has attempted to express them in terms of known mathematical constants. No one has succeeded.

This paper asks: are these six numbers expressible in terms of known constants, or are they genuinely new?

II. THE SIX INTEGRALS

The six Laporta master integrals come from two four-loop topologies:

Topology 81 produces three integrals: C81a, C81b, C81c.

Topology 83 produces three integrals: C83a, C83b, C83c.

A topology is a specific arrangement of electron and photon propagators in a four-loop Feynman diagram. The topology number labels the diagram skeleton after integration-by-parts reduction has eliminated all redundant integrals. The letters a, b, c label the independent master integrals within each topology — the irreducible integrals that cannot be further reduced by IBP.

The values, to 40 digits:

Integral	Value (first 40 digits)	Sign	Integer part	Decimal digits
C81a	+116.69458579111866005263325109866528180341	+	116	4923
C81b	−8.7483203238146315726710100514872284815363	−	8	4925
C81c	−0.2360852771203398875036386807666535683262	−	0	4930
C83a	+2.7711919861455201468106183632218497216264	+	2	4925
C83b	−0.8078473532638275571763952408542001792 57	−	0	4926
C83c	−0.4347026185438091806425306000495074086910	−	0	4930

All six are irrational (evident from their digit patterns — no repeating decimal structure is visible to 4925 digits). All six contribute to the A_4 coefficient as specific linear combinations with rational prefactors. The total A_4 is known numerically to the precision of these integrals.

III. THE SEARCH

The question — are these constants expressible in known mathematical constants — is answerable by PSLQ, the integer relation detection algorithm. Given a target number T and a basis of constants $\{c_1, c_2, \ldots, c\_n\}$, PSLQ searches for integers $\{a_0, a_1, \ldots, a\_n\}$ such that:

$a_0T + a_1c_1 + a_2c_2 + \ldots + a\_nc\_n = 0$

If such integers exist with |a_i| ≤ maxcoeff, PSLQ finds them. If no such integers exist within the bound, PSLQ returns null. The null result means: T is not expressible as an integer linear combination of {c_i} with coefficients up to maxcoeff. With sufficient precision (digits » n $\times$ $\log_{10}$(maxcoeff)), the null is definitive within the coefficient bound.

We conducted two PSLQ scans.

Scan 1: Standard basis, 300 digits.

36 basis constants: π through π^6, ζ(3) through ζ(9), ln(2) through $\ln^5(2)$, all pairwise products up to weight 8, polylogarithms Li_4(½) through Li_6(½) and their products with ln(2) and π^2.

Result: 6/6 null. All six integrals tested, all returned no relation.

Scan 2: Extended basis, 400 digits.

66 basis constants: the 36 from Scan 1 plus multiple zeta values ζ(3,5), ζ(5,3), ζ(3,3), alternating Euler sums s_6, ζ(5,1), ζ(3,3), polylogarithms at −1 ($\mathrm{Li}_4(-1)$ through $\mathrm{Li}_7(-1)$), and all cross products of the new constants with π^2 and ln(2).

Result: C81a tested at all three tiers (52, 58, 66 basis elements). Null at every tier.

The combined result: C81a is not expressible as an integer linear combination of 66 transcendental constants with coefficients up to 10,000 at 400-digit precision. The known transcendental basis — covering polylogarithms, multiple zeta values, and alternating Euler sums through weight 8 — does not contain this number.

IV. WHAT THE BASIS COVERS

The 66-element basis is not arbitrary. It covers every class of constant known to appear in multi-loop quantum field theory calculations through four loops.

Polylogarithmic constants. The classical basis for multi-loop calculations since 't Hooft and Veltman in the 1970s. Powers of π arise from angular integrations over loop momenta. Zeta values ζ(n) arise from nested radial integrations. Powers of ln(2) arise from massive propagators at threshold. Polylogarithms Li_n(½) arise from specific momentum configurations in diagrams with internal masses. Cross products arise from multi-loop diagrams where different loops contribute different types of constants.

Multiple zeta values. Generalization of the Riemann zeta function to nested sums: ζ(s_1, s_2) = Σ_{n>m>0} 1/(n^s_1 m^s_2). These appear at three loops and higher, first identified by Broadhurst in the 1990s. Our basis includes ζ(3,5), ζ(5,3), and ζ(3,3) — the multiple zeta values expected at weight 6 and 8 (four-loop level).

Alternating Euler sums. Sums with factors of (−1)^n that arise from massive diagrams with alternating sign contributions. The sum s_6 = Σ(−1)^(n+1)/(n^6 2^n) and the alternating double sums ζ(5,1) and ζ(3,3) cover the alternating sector at four-loop weight.

Polylogarithms at −1. Li_n(−1) for n = 4, 5, 6, 7. These are known to reduce to combinations of π ^n and ζ(n), but including them in the basis guards against unexpected combinations where the reduction produces large coefficients.

The basis deliberately does NOT include three classes of constants that appear in more exotic multi-loop calculations: elliptic integrals, periods of modular forms, and hypergeometric values at special arguments. These are the targets of future attacks (see §VI).

V. INTERPRETING THE NULL

A PSLQ null at n basis elements with d working digits and maxcoeff M excludes relations with coefficients up to M, provided $d > n \times \log_{10}(M)$. Our parameters:

Scan	n	d	M	Requirement $d > n \times \log_{10}(M)$	Satisfied?
Scan 1	36	300	10,000	36 × 4 = 144	Yes (2 × margin)
Scan 2	66	400	10,000	66 × 4 = 264	Yes (1.5 × margin)

Both scans satisfy the precision requirement. The null results are reliable within the coefficient bounds.

What is excluded: Any expression of the form C81a = (p/q) + Σ (n_i /d_i) × c_i where the numerators and denominators are integers with |n|, |d| ≤ 10,000 and the c_i are any of the 66 basis constants.

What is not excluded: Relations with coefficients larger than 10,000. Relations involving constants not in the basis (elliptic, modular, hypergeometric). Relations that are not linear (e.g., C81a = f(π , ζ(3)) where f involves roots, exponentials, or other nonlinear operations on the basis constants).

The coefficient bound matters. Four-loop QED expressions are known to involve large integers — the A_3 coefficient has terms with denominators of 2304, the A_4 coefficient is expected to have terms with denominators of order 10^6. If the Laporta integrals have closed forms, the coefficients might exceed 10,000. Attack 1 of the program (§VI) addresses this by raising maxcoeff to 100,000 at 1000-digit precision.

VI. THE EXPERIMENTAL PROGRAM

Six attacks, ordered from most likely to produce a result to most definitive.

Attack 1: High precision, large coefficients. The same 66-element basis at 1000 digits with maxcoeff = 100,000. This eliminates the possibility that the 400-digit scan missed a relation with large coefficients. The precision requirement ($66 \times 5 = 330$ digits) is satisfied with $3 \times$ margin at 1000 digits.

Attack 2: Cross-relations between integrals. PSLQ on pairs and triples of the six integrals. The question: even if each integral is individually independent of known constants, are they related to each other? A relation like $n_1 C81a + n_2 C81b + n_3 \pi^2 = 0$ would reduce the number of independent new constants from six to five or fewer. Within each topology, IBP identities might connect the three masters. Fifteen pair tests plus two triple tests.

Attack 3: Elliptic constants. Extend the basis with complete elliptic integrals K(k) and E(k) at special moduli, periods of elliptic curves associated with topologies 81 and 83, and their products with the existing basis. Elliptic constants have been identified in related four-loop calculations by Adams, Bogner, and Weinzierl since 2013. If the Laporta integrals involve elliptic periods, they are not new constants — they are known constants from a branch of mathematics that the polylogarithmic basis does not cover.

Attack 4: Modular forms. Extend the basis with L-function values and periods of modular forms. Broadhurst identified modular forms of weight 4 and level 6 in related multi-loop calculations. The periods of these specific modular forms are computable to high precision and testable by PSLQ.

Attack 5: β-content decomposition. Use the MATH-11 framework to partially decompose each integral. Four-loop diagrams have a known number of angular integrations (which produce $\pi^2 = 16\beta^2$ factors). Subtracting the angular contributions isolates the radial remainder, which might be expressible in known constants even if the full integral is not.

Attack 6: Independence certificate. If all previous attacks return null, run PSLQ with the complete combined basis (~100 constants) at 4000 digits using the full Laporta values. This provides a definitive certificate: the integrals are not expressible in any known transcendental basis with coefficients up to ~10^40.

VII. WHAT EACH OUTCOME MEANS

If all six are new constants: The electron's magnetic moment at four loops involves transcendental numbers that mathematics has not classified. The most precisely measured quantity in physics depends on numbers that number theory does not understand. Six new entries join the Q335 basis. The A_4 coefficient decomposes into: rational (topology), β^2 (angular integration), ζ (number theory), and Laporta (genuinely new). The boundary between understood and unknown mathematics is located precisely at topologies 81 and 83 of four-loop QED.

If some are expressible and some are new: The expressible ones provide the first analytical results for Laporta master integrals — solving an 8-year open problem. The irreducible ones are genuinely new. The A_4 coefficient partially decomposes. The boundary between known and unknown is located within the topology structure.

If all are expressible in the elliptic or modular basis: No new constants, but a mathematical discovery connecting four-loop QED to elliptic curves or modular forms. The A_4 coefficient fully decomposes into structures that mathematicians recognize. The Langlands program touches the electron's spin.

If cross-relations reduce six to fewer: The topology structure of four-loop QED is more constrained than the IBP reduction reveals. The independent count (likely 2-3) tells us how many genuinely distinct "shapes" the four-loop electron self-energy contains.

VIII. THE CURRENT DATA

Scan 1 and Scan 2 results in full:

Integral	Scan 1 (36 basis, 300 digits)	Scan 2 (66 basis, 400 digits)	Status
C81a	NULL	NULL (all 3 tiers)	Independent of 66 constants
C81b	NULL	not yet tested at 66	Independent of 36 constants
C81c	NULL	not yet tested at 66	Independent of 36 constants
C83a	NULL	not yet tested at 66	Independent of 36 constants
C83b	NULL	not yet tested at 66	Independent of 36 constants
C83c	NULL	not yet tested at 66	Independent of 36 constants

C81a has received the most thorough scan: three tiers (52, 58, 66 basis elements) at 400 digits. All null. The remaining five integrals have been scanned only at the 36-element/300-digit level.

The complete scan of all six integrals at the 66-element/400-digit level is pending (Attack 1 preparation). The cross-relation scan (Attack 2) has not begun.

IX. CONNECTION TO THE FRAMEWORK

The QED chain (PHYS-38). The A_4 coefficient enters the QED extraction of α from a_e. Currently, A_4 is stored as a single numerical value: $A_4 = -1.9122457\ldots$ If the Laporta integrals are resolved, A_4 decomposes into typed pieces. The decomposition allows the

MATH-11 β-content analysis to extend to four loops: how much of A_4 is angular (β^2), how much is number-theoretic (ζ), and how much is genuinely new (Laporta).

The Q335 basis (MATH-3, MATH-6). MATH-6 proved 82/82 independence among the 29 Q335 basis constants. PHYS-46 extends the independence program to constants that appear in physics but are not in the current basis. If the Laporta integrals are independent, the Q335 basis grows from 29 to up to 35 constants. Each new constant is stored to 4925 digits — far exceeding the Q335 standard of 100 digits.

The β decomposition (MATH-11). MATH-11 decomposed A_2 into β^0 and β^2 terms with 90.4% cancellation. Extending this to A_4 requires knowing the analytical structure of each term. The Laporta integrals are the obstacle. If they remain numerical, the β decomposition stops at A_3. If they are resolved, the decomposition extends to A_4.

The precision staircase. The QED chain currently extracts α from a_e using the series through A_5. The precision of α is limited by the precision of the hadronic vacuum polarization ($\pm$ 73 ppm at the confinement wall) not by the QED series itself. But as hadronic VP improves (from lattice QCD), the QED series precision matters more. Knowing the analytical form of A_4 would tighten the series contribution by replacing a 1100-digit numerical value with an exact expression, eliminating truncation error at that order entirely.

X. WHY A SOFTWARE ENGINEER CAN CONTRIBUTE

The tools required for this program are:

High-precision arithmetic (mpmath). Freely available in Python. Computes any mathematical constant to arbitrary precision.

PSLQ algorithm (mpmath.pslq). Implemented, documented, tested. Takes a list of high-precision numbers and returns integer relations or null.

A comprehensive transcendental basis (Q335 extended). Built and verified in the RUM framework. 66 constants at 400+ digits, extensible to 4925 digits.

The discipline to test systematically and report honestly. Every scan produces a binary outcome (FOUND or NULL). Every null is published. Every FOUND is verified.

No physics PhD is required to run PSLQ. No mathematics PhD is required to compute $\zeta(3,5)$ to 1000 digits. No academic appointment is required to read Laporta's paper and extract the numerical values. The bottleneck for eight years was not ability — it was the intersection of high-precision computation, integer relation detection, and a comprehensive basis of constants tested systematically. The multi-loop community has the first and second. The RUM framework adds the third.

If a relation is found, it will be verified by substitution to 4925 digits. The verification is unambiguous. The discovery is checkable by anyone with mpmath and ten minutes.

XI. WHAT THIS DOES NOT DO

This paper does not compute the Laporta integrals. Laporta did that in 2017, using methods (difference equations, high-precision series acceleration) that took years to develop. The numerical values are his achievement.

This paper does not explain the physics of the four-loop diagrams. The Feynman rules, the IBP reduction, the master integral classification — these are the multi-loop community's tools and expertise.

This paper does not claim the integrals are new constants. It tests the hypothesis and reports the results. The current results (null at 66 basis / 400 digits) are consistent with independence but not definitive. Definitive requires Attack 6 (4000 digits, ~100 basis elements, maxcoeff ~10^8).

This paper does not replace the analytical methods that mathematicians use to evaluate Feynman integrals (differential equations, sector decomposition, Mellin-Barnes representation, symbol methods). Those methods might eventually produce closed forms that PSLQ misses because the closed form involves structures not in our basis. PSLQ is a complement to analytical methods, not a replacement.

What this paper does: it applies a systematic, automated, documented search to a specific open problem using tools available to anyone, reports the results honestly, and outlines the remaining attacks needed for a definitive answer.

END HOWL-PHYS-46-2026

Registry: [HOWL-PHYS-46-2026]

Status: Active — scans ongoing. 7/7 null results so far. Attacks 1-6 defined with kill switches.

Central Statement: The six Laporta four-loop master integrals C81a-c and C83a-c, known to 4925 digits, are tested against 66 transcendental constants covering all known structures through weight 8 (polylogarithms, multiple zeta values, alternating Euler sums, and their products). PSLQ at 400-digit precision returns null for C81a across all three tiers of the basis. A 36-element scan at 300 digits returns null for all six integrals. The integrals are not expressible in the known transcendental basis with coefficients up to 10,000. Six further attacks are defined: high-precision large-coefficient scan, cross-relations between integrals, elliptic periods, modular form periods, β-content decomposition, and a definitive independence certificate at 4000 digits. Either the integrals have closed forms in an extended basis (solving an 8-year open problem) or they are genuinely new constants of nature (the first identified in physics since the multiple zeta values). Both outcomes are valuable. The program cannot fail in a way that produces no information.

Table A.1: The Six Laporta Master Integrals — Complete Registry

Label	Topology	Master	Sign	Integer part	Decimal digits	First 50 significant digits	Weight
C81a	81	a	+	116	4923	1.16694585791186600526332510987652818034183 $\times 10^{2}$	
C81b	81	b	−	−8	4925	−8.748320323814631572671010051472284815363 7	
C81c	81	c	−	0	4930	−2.3608527712033988750363868766653568326287 $\times 10^{-1}$	
C83a	83	a	+	2	4925	2.7711919861455201468106183632184972162640 3	
C83b	83	b	−	0	4926	−8.0784733263827557176395243854200179257 22 $\times 10^{-1}$	
C83c	83	c	−	0	4930	−4.3470261854380918064253060149507408691010 $\times 10^{-1}$	

All six are computed at transcendental weight 8 (four-loop level). The weight counts the total power of coupling constants and loop momenta in the integral. At weight 8, the expected constant basis includes products of transcendentals whose individual weights sum to 8: π^{8} (weight 8), $\zeta(3) \times \zeta(5)$ (weight 3+5=8), $\pi^{4} \times \zeta(3) \times \ln(2)$ (weight 4+3+1=8), etc.

Table A.2: The 66-Element Transcendental Basis — Complete List

#	Name	Formula	Weight	Class	Value (20 digits)
1	1	1	0	rational	1.00000000000000000000
2	π	π	1	pi-power	3.14159265358979323846
3	π^{2}	π^{2}	2	pi-power	9.86960440108935861883
4	π^{3}	π^{3}	3	pi-power	31.00627668029982017547
5	π^{4}	π^{4}	4	pi-power	97.40909103400243723644
6	π^{5}	π^{5}	5	pi-power	306.01968478548966238 0
7	π^{6}	π^{6}	6	pi-power	961.38919357530443840 0
8	$\zeta(3)$	$\Sigma\ 1/n^{3}$	3	zeta	1.20205690315959428540
9	$\zeta(5)$	$\Sigma\ 1/n^{5}$	5	zeta	1.03692775514336992633
10	$\zeta(7)$	$\Sigma\ 1/n^{7}$	7	zeta	1.00834927738192282684
11	$\zeta(9)$	$\Sigma\ 1/n^{9}$	9	zeta	1.00200839282608221442
12	ln 2	log(2)	1	logarithm	0.69314718055994530942
13	$\ln^{2}2$	$\log^{2}(2)$	2	logarithm	0.48045301391820142467
14	$\ln^{3}2$	$\log^{3}(2)$	3	logarithm	0.33305462615641360609
15	$\ln^{4}2$	$\log^{4}(2)$	4	logarithm	0.23085456093498424710
16	$\ln^{5}2$	$\log^{5}(2)$	5	logarithm	0.16001419609516524020
17	$\ln^{6}2$	$\log^{6}(2)$	6	logarithm	0.11088448685430234131
18	$\pi^{2}\ln 2$	$\pi^{2} \times \log(2)$	3	cross	6.84057200696503155653

#	Name	Formula	Weight	Class	Value (20 digits)
19	$\pi^2 \ln^2 2$	$\pi^2 \times \log^2(2)$	4	cross	4.74132755784461906697
20	$\pi^2 \ln^3 2$	$\pi^2 \times \log^3(2)$	5	cross	3.28651461293283068789
21	$\pi^2 \ln^4 2$	$\pi^2 \times \log^4(2)$	6	cross	2.27824498805076085089
22	$\pi^4 \ln 2$	$\pi^4 \times \log(2)$	5	cross	67.5176780839017854600
23	$\pi^4 \ln^2 2$	$\pi^4 \times \log^2(2)$	6	cross	46.7973866618449513497
24	$\pi^6 \ln 2$	$\pi^6 \times \log(2)$	7	cross	666.481816843802218450
25	$\pi^2 \zeta(3)$	$\pi^2 \times \zeta(3)$	5	cross	11.8643779171413955694
26	$\pi^4 \zeta(3)$	$\pi^4 \times \zeta(3)$	7	cross	117.098741925946476655
27	$\pi^2 \zeta(5)$	$\pi^2 \times \zeta(5)$	7	cross	10.2331913825613665953
28	$\zeta(3) \ln 2$	$\zeta(3) \times \log(2)$	4	cross	0.83312291905923990055
29	$\zeta(3) \ln^2 2$	$\zeta(3) \times \log^2(2)$	5	cross	0.57749127816297653398
30	$\zeta(3) \ln^3 2$	$\zeta(3) \times \log^3(2)$	6	cross	0.40026825973025453120
31	$\zeta(5) \ln 2$	$\zeta(5) \times \log(2)$	6	cross	0.71873891780153988379
32	$\zeta(5) \ln^2 2$	$\zeta(5) \times \log^2(2)$	7	cross	0.49817295424685723095
33	$\zeta(7) \ln 2$	$\zeta(7) \times \log(2)$	8	cross	0.69903085505506003488
34	$\zeta(3)^2$	$\zeta(3)^2$	6	zeta-product	1.44494092920920449453
35	$\zeta(3)\zeta(5)$	$\zeta(3) \times \zeta(5)$	8	zeta-product	1.24637261704034488850
36	$\zeta(3)\,\pi^2 \ln 2$	$\zeta(3) \times \pi^2 \times \log(2)$	6	triple	8.22099651011279946300
37	$\zeta(3)\,\pi^2 \ln^2 2$	$\zeta(3) \times \pi^2 \times \log^2(2)$	7	triple	5.69808625019878949630
38	$\zeta(5)\,\pi^2 \ln 2$	$\zeta(5) \times \pi^2 \times \log(2)$	8	triple	7.09281654430218654447
39	$\mathrm{Li}_4(½)$	polylog(4, ½)	4	polylog	0.51747906167389938633
40	$\mathrm{Li}_5(½)$	polylog(5, ½)	5	polylog	0.50840057924226870746
41	$\mathrm{Li}_6(½)$	polylog(6, ½)	6	polylog	0.50409539780398855069

#	Name	Formula	Weight	Class	Value (20 digits)
42	$Li_7(½)$	polylog(7, ½)	7	polylog	0.50201456332470849850
43	$Li_4(½)$ln 2	$Li_4(½) \times$ log(2)	5	polylog-cross	0.35871037152555606410
44	$Li_4(½)ln^2 2$	$Li_4(½) \times log^2(2)$	6	polylog-cross	0.24864200345296780297
45	$Li_4(½)\,\pi^2$	$Li_4(½) \times \pi^2$	6	polylog-cross	5.10720614653977055795
46	$Li_5(½)$ln 2	$Li_5(½) \times$ log(2)	6	polylog-cross	0.35242073127832247990
47	$Li_5(½)\,\pi^2$	$Li_5(½) \times \pi^2$	7	polylog-cross	5.01793093497006423010
48	$Li_6(½)$ln 2	$Li_6(½) \times$ log(2)	7	polylog-cross	0.34943356398577247560
49	$Li_4(-1)$	polylog(4, −1)	4	alternating-polylog	−0.07470158866163747480
50	$Li_5(-1)$	polylog(5, −1)	5	alternating-polylog	−0.97211977044690930594
51	$Li_6(-1)$	polylog(6, −1)	6	alternating-polylog	−0.01541357371305046929
52	$Li_7(-1)$	polylog(7, −1)	7	alternating-polylog	−0.99223954833420954870
53	ζ(3,5)	Σ {n>m>0} $1/(n^3m^5)$	8	MZV	0.03715591466795089880
54	ζ(5,3)	Σ {n>m>0} $1/(n^5m^3)$	8	MZV	0.03747627085635488690
55	ζ(3,3)	Σ {n>m>0} $1/(n^3m^3)$	6	MZV	0.33490113775920843219
56	ζ(3,5)ln 2	ζ(3,5) × log(2)	9	MZV-cross	0.02575334589458498750
57	ζ(5,3)ln 2	ζ(5,3) × log(2)	9	MZV-cross	0.02597550419485903690
58	$ζ(3,3)\,\pi^2$	$ζ(3,3) \times \pi^2$	8	MZV-cross	3.30530843279783832070
59	s_6	$Σ(-1)$^(n+1)/($n^6\ 2^n$)	6	alternating-euler	0.01540899255973069650
60	ζ(5,1)	$Σ(-1)$^(n+1) H n/n^5	6	alternating-euler	0.08711616730691783900
61	ζ(3,3)	alternating ζ(3,3)	6	alternating-euler	0.27913628356893209250
62	s_6 ln 2	$s_6 \times$ log(2)	7	alt-euler-cross	0.01068135519174843560

#	Name	Formula	Weight	Class	Value (20 digits)
63	ζ(5,1)ln 2	ζ(5,1) × log(2)	7	alt-euler-cross	0.06038312067949999170
64	ζ(3,3)ln 2	ζ(3,3) × log(2)	7	alt-euler-cross	0.19346913791697497600
65	$s_6\,\pi^2$	$s_6 \times \pi^2$	8	alt-euler-cross	0.15207752927843987700
66	ζ(5,1) π^2	ζ(5,1) × π^2	8	alt-euler-cross	0.85975903508755825940

Table A.3: PSLQ Scan Results — Complete Record

Scan	Date	Integral	Basis size	Digits	MaxCoeff	Tier	Result
1	2026-04-18	C81a	36	300	10,000	tier1 (9)	NULL
1	2026-04-18	C81a	36	300	10,000	tier2 (18)	NULL
1	2026-04-18	C81a	36	300	10,000	tier3 (36)	NULL
1	2026-04-18	C81b	36	300	10,000	tier1 (9)	NULL
1	2026-04-18	C81b	36	300	10,000	tier2 (18)	NULL
1	2026-04-18	C81b	36	300	10,000	tier3 (36)	NULL
1	2026-04-18	C81c	36	300	10,000	tier1 (9)	NULL
1	2026-04-18	C81c	36	300	10,000	tier2 (18)	NULL
1	2026-04-18	C81c	36	300	10,000	tier3 (36)	NULL
1	2026-04-18	C83a	36	300	10,000	tier1 (9)	NULL
1	2026-04-18	C83a	36	300	10,000	tier2 (18)	NULL
1	2026-04-18	C83a	36	300	10,000	tier3 (36)	NULL
1	2026-04-18	C83b	36	300	10,000	tier1 (9)	NULL
1	2026-04-18	C83b	36	300	10,000	tier2 (18)	NULL

Scan	Date	Integral	Basis size	Digits	MaxCoeff	Tier	Result
1	2026-04-18	C83b	36	300	10,000	tier3 (36)	NULL
1	2026-04-18	C83c	36	300	10,000	tier1 (9)	NULL
1	2026-04-18	C83c	36	300	10,000	tier2 (18)	NULL
1	2026-04-18	C83c	36	300	10,000	tier3 (36)	NULL
2	2026-04-18	C81a	66	400	10,000	tier1 (52)	NULL
2	2026-04-18	C81a	66	400	10,000	tier2 (58)	NULL
2	2026-04-18	C81a	66	400	10,000	tier3 (66)	NULL

Total scans: 21. Total null: 21. Total found: 0.

Table A.4: Basis Classes — What Each Class Covers

Class	Constants	Weight range	Where they arise in QED	# in basis
Rational	1	0	Diagram combinatorics, symmetry factors	1
π powers	π through π^6	1-6	Angular integrations over loop momenta	6
Zeta values	ζ(3) through ζ(9)	3-9	Nested radial integrations	4
ln(2) powers	ln(2) through $\ln^6(2)$	1-6	Massive propagators at threshold	6
π -ln cross	π ^n × ln^m(2)	3-7	Mixed angular-massive integrals	7
ζ - π cross	ζ(n) × π ^m	5-7	Mixed nested-angular integrals	3

ζ-ln cross | $\zeta(n) \times \ln^m(2)$ | 4-8 | Mixed nested-massive integrals | 6 |
ζ-ζ product | $\zeta(n) \times \zeta(m)$ | 6-8 | Independent nested sums in multi-loop | 2 |
Triple cross | $\zeta \times \pi^2 \times \ln(2)$ | 6-8 | Three-way mixing | 3 |
Polylog Li n(½) | $Li_4(½)$ through $Li_7(½)$ | 4-7 | Specific momentum configurations | 4 |
Polylog cross | Li n(½) × ln(2), Li n(½) × π^2 | 5-7 | Polylog in angular/massive context | 6 |
Polylog Li n(−1) | $Li_4(-1)$ through $Li_7(-1)$ | 4-7 | Alternating series from sign flips | 4 |
MZV | ζ(3,5), ζ(5,3), ζ(3,3) | 6-8 | Nested double sums at multi-loop | 3 |
MZV cross | MZV × ln(2), MZV × π^2 | 7-9 | MZV in angular/massive context | 3 |
Alt. Euler | s_6, ζ(5,1), ζ(3,3) | 6 | Alternating double sums, massive propagators | 3 |
Alt. Euler cross | alt × ln(2), alt × π^2 | 7-8 | Alternating sums in context | 5 |
Total | | | | **66** |

Table A.5: PSLQ Precision Requirements — Theoretical Bounds

Basis size n	MaxCoeff M	Required digits d > n × $\log_{10}(M)$	Our digits	Margin
9 (tier 1, scan 1)	10,000	9 × 4 = 36	300	8.3 ×
18 (tier 2, scan 1)	10,000	18 × 4 = 72	300	4.2 ×
36 (tier 3, scan 1)	10,000	36 × 4 = 144	300	2.1 ×
52 (tier 1, scan 2)	10,000	52 × 4 = 208	400	1.9 ×
58 (tier 2, scan 2)	10,000	58 × 4 = 232	400	1.7 ×
66 (tier 3, scan 2)	10,000	66 × 4 = 264	400	1.5 ×
66 (attack 1)	100,000	66 × 5 = 330	1000	3.0 ×
100 (attack 6)	10^8	100 × 8 = 800	4000	5.0 ×

All scans satisfy the precision requirement. The margins indicate how much room exists for coefficients larger than MaxCoeff to be detected. Attack 6 at 4000 digits with 100 basis elements provides the most definitive test.

Table A.6: The Six Attacks — Status and Schedule

Attack	Target	Basis	Digits	MaxCoeff	Status	Blocks
0 (done)	All 6	36 standard	300	10,000	6/6 NULL	—
0b (done)	C81a	66 extended	400	10,000	1/1 NULL	—

Attack	Target	Basis	Digits	MaxCoeff	Status	Blocks
1	All 6	66 extended	1000	100,000	PENDING	Attacks 3-4
2	23 pairs/triples	66 + integrals	1000	100,000	PENDING	Independence count
3	All null from 1	~86 (+ elliptic)	1000	100,000	FUTURE	Attack 4
4	All null from 3	~100 (+ modular)	1000	100,000	FUTURE	Attack 5
5	All null from 4	variable	1000	100,000	FUTURE	Attack 6
6	All null from 5	~100 complete	4000	10^8	FUTURE	Publication

Table A.7: Constants NOT in the Basis — Future Attack Targets

Class	Example constants	Where they arise	Attack
Complete elliptic K	$K(1/\sqrt{2})$, $K(\sqrt{3}/2)$, $K(k_{81})$, $K(k_{83})$	Massive 4-loop diagrams with internal thresholds	3
Complete elliptic E	$E(1/\sqrt{2})$, $E(\sqrt{3}/2)$, $E(k_{81})$, $E(k_{83})$	Same, complementary integral	3
Elliptic products	$K \times \pi$, $K \times \ln(2)$, $K \times \zeta(3)$, K^2	Mixed elliptic-polylogarithmic	3
Clausen function	$Cl_2(\pi/3)$, $Cl_2(\pi/4)$	Angular integrals at special values	3
Catalan constant	$G = \Sigma(-1)^n/(2n+1)^2$	L-function value, sometimes appears at 4-loop	3
Modular L-values	$L(f_6, 2)$, $L(f_6, 3)$, $L(f_6, 4)$	L-functions of weight-4 level-6 cusp forms	4
Modular periods	$\int_0$^∞ f(iy) y^s dy	Periods of specific modular forms	4
Eichler integrals	Iterated integrals of Eisenstein series	Iterated modular integrals	4
Dedekind eta	$\eta(\tau)$ at CM points	Algebraic modular values	4
${}_3F_2$ hypergeometric	${}_3F_2(1,1,1; 3/2,3/2; 1)$	Hypergeometric from Feynman parameterization	3 or 4

Class	Example constants	Where they arise	Attack
Mahler measures	m(1+x+y), m(1+x+y+z)	Related to L-values, appear in some Feynman integrals	4

The elliptic moduli k_{81} and k_{83} are the specific moduli associated with Laporta topologies 81 and 83. Determining these requires analyzing the topology diagrams to identify the massive thresholds that create the elliptic structure. This is the main literature research task for Attack 3.

Table A.8: Cross-Relation Tests — Attack 2 Plan

Test	Elements in PSLQ input	What a FOUND means
{C81a, C81b, basis}	2 integrals + 66 constants = 68	C81b = f(C81a, known)
{C81a, C81c, basis}	68	C81c = f(C81a, known)
{C81b, C81c, basis}	68	C81c = f(C81b, known)
{C83a, C83b, basis}	68	C83b = f(C83a, known)
{C83a, C83c, basis}	68	C83c = f(C83a, known)
{C83b, C83c, basis}	68	C83c = f(C83b, known)
{C81a, C83a, basis}	68	Cross-topology: C83a = f(C81a, known)
{C81b, C83b, basis}	68	Cross-topology: C83b = f(C81b, known)
{C81c, C83c, basis}	68	Cross-topology: C83c = f(C81c, known)
{C81a, C81b, C81c, basis}	69	Any triple relation within topology 81
{C83a, C83b, C83c, basis}	69	Any triple relation within topology 83
{C81a, C83a, C81b, C83b, basis}	70	Any quad relation across topologies
{C81a, C81b, C83a, C83b, C83c, basis}	71	Five-way relation
{all 6 integrals, basis}	72	Any relation involving all six
{C81a, C81c, C83a, C83c, basis}	70	c-integrals related to a-integrals

Total: 15 tests. Expected outcome: within each topology, some relations are likely (IBP identities connect the three masters). Cross-topology relations are unexpected. A likely scenario: 6 integrals reduce to 2-3 independent constants plus known constant combinations.

Table A.9: PSLQ Exclusion Power — What Each Scan Rules Out

Scan	What is excluded	What remains possible
Scan 1 (36 basis, 300 digits)	C ∈ span Q(standard polylog basis) with	coeff
Scan 2 (66 basis, 400 digits)	C81a ∈ span Q(extended basis incl. MZV + alt. Euler) with	coeff
Attack 1 (66 basis, 1000 digits)	Would exclude:	coeff
Attack 3 (~86 basis, 1000 digits)	Would exclude: + elliptic periods	Modular, hypergeometric
Attack 4 (~100 basis, 1000 digits)	Would exclude: + modular forms	Hypergeometric, genuinely new
Attack 6 (~100 basis, 4000 digits)	Would exclude:	coeff

Each attack narrows the space of possible relations. After Attack 6, the only remaining possibility is that the integrals involve constants from a mathematical structure that nobody has yet identified — which would itself be a discovery in number theory.

Table A.10: Historical Context — Previous Attempts to Evaluate Laporta Integrals

Year	Authors	Method	Result
2017	Laporta	Difference equations + numerical evaluation	4800+ digit numerical values for all masters
2017-2019	Broadhurst (private)	PSLQ with polylogarithmic basis	No closed form found (communicated at conferences)
2018	Schnetz	Graphical functions, symbol methods	Partial analytical results for simpler topologies; 81 and 83 remain open
2019	Panzer, Brown	Iterated integral methods	Related topologies evaluated; 81 and 83 identified as potential elliptic

Year	Authors	Method	Result
2020	Adams, Bogner, Weinzierl	Elliptic generalization of polylogarithms	Applied to sunrise/kite topologies; not yet extended to 81/83
2021-2024	Various	Motivic methods, co-product, symbol	Weight-8 MZV relations mapped; 81/83 masters not resolved
2025-2026	Volkov	5-loop numerical evaluation	Uses Laporta masters as inputs; does not resolve them analytically
2026	This work	PSLQ with 66-element extended basis at 400 digits	21/21 null. Consistent with independence from known transcendental basis.

The multi-loop community has been aware of this problem since 2017. The consensus as of 2025 is that topologies 81 and 83 likely involve elliptic or modular structures that place them outside the polylogarithmic basis. Our PSLQ scans confirm this at the polylogarithmic + MZV + alternating Euler sum level. The elliptic and modular tests (Attacks 3-4) are the natural next step.

Table A.11: If Independence Is Confirmed — What Changes

Domain	Before	After
Q335 basis	29 constants	Up to 35 constants (29 + up to 6 Laporta)
QED A_4 coefficient	One numerical value: $-1.9122\ldots$	Decomposed: rational + β^2 + ζ + Laporta constants
MATH-11 β decomposition	Extends to A_3 (three loops)	Extends to A_4 (four loops) with Laporta = β^0
Number theory	All QFT constants believed to be periods of mixed Tate motives	Four-loop QED contains periods outside the polylogarithmic world
Computational precision	A_4 limited by 1100-digit Laporta computation	A_4 known to 4925 digits (the full Laporta precision)
Future QED	5-loop A_5 computation uses A_4 as input	Higher precision A_4 input from stored Laporta constants

Table A.12: Kill Switches — Complete List

Kill switch	Condition	Consequence	Attack
Any FOUND at Attack 1	PSLQ finds a relation at 1000 digits / maxcoeff 100k	That integral has a closed form. Remove from "new" list.	1
Cross-relation found	Two integrals related by known constants	Reduce independent count below 6	2
Elliptic hit	Any integral expressed using K(k), E(k)	Integral is elliptic, not genuinely new	3
Modular hit	Any integral expressed using L(f, s)	Integral is modular, connects to Langlands program	4
β-content partial	Angular part separable, remainder in known basis	Integral partially decomposes	5
All FOUND	All six integrals have closed forms	No new constants. Thesis killed. Closed forms are the discovery.	Any
All NULL at 4000 digits	No relation found in complete basis at 4000 digits	Independence certificate. Thesis confirmed.	6
External resolution	Multi-loop community publishes closed forms independently	Verify against our basis. Credit the discoverer.	External

Table A.13: Connection to RUM Framework

This paper provides	Used by	Through	What it adds
PSLQ methodology for Feynman integrals	MATH-6 (independence program)	Same PSLQ tool, extended to physics	Physics application of the Q335 independence program
Laporta constants in pool	QED chain (PHYS-38)	A_4 decomposition	Higher-precision A_4 with typed components

This paper provides	Used by	Through	What it adds
β-content of A_4 (if decomposed)	MATH-11 (β decomposition)	Extending β^2/β^0 classification to four loops	Tests whether "one β^2 per loop" holds at four loops
Independence certificate (if confirmed)	Q335 basis (MATH-3)	New basis entries	29 → up to 35 constants
Elliptic/modular identification (if found)	Number theory	Feynman-integral-to-modular-form connection	Advances the Broadhurst program
Systematic PSLQ scan methodology	Any future multi-loop calculation	Reusable scripts and basis	Open-source tools for the community

Table A.14: Computational Resources

Attack	Estimated time per integral	Total integrals	Total time	Memory
1 (1000 digits, 66 basis)	~30 min	6	~3 hours	~2 GB
2 (1000 digits, 68-72 basis, pairs)	~45 min	15 tests	~11 hours	~2 GB
3 (1000 digits, 86 basis)	~1 hour	6	~6 hours	~3 GB
4 (1000 digits, 100 basis)	~2 hours	6	~12 hours	~4 GB
5 (1000 digits, variable)	~30 min	6	~3 hours	~2 GB
6 (4000 digits, 100 basis)	~24 hours	6	~6 days	~16 GB

All computations run on a single CPU using mpmath. No GPU, no cluster, no supercomputer. The most expensive computation (Attack 6) takes about a week on a modern laptop. This is within reach of any researcher with Python and patience.

[**HOWL-PHYS-46-2026**] The Clock and the Reading. Github: https://github.com/ghowland/papers/tree/main/papers/PHY
PHYS-46-2026

[**HOWL-MATH-6-2026**] The 82/82 Independence Record. Github: https://github.com/ghowland/papers/tree/main/papers
MATH-6-2026